COMPETITIVE RANKING REPORT · FIVE GUARDRAIL ARMS · TWO CORPORA

PANOPTES vs the Field

A measured, head-to-head ranking of the PANOPTES prototype against the real OpenAPPA v0.30.0 binary, an APPA blunt-patch variant, a regex/DLP blocklist, and an LLM-as-judge — on our 40-scenario corpus and on OpenAPPA's own corp, AgentThreatBench, and τ-bench harnesses with real LLM agents. Every number is a measurement; provenance is labeled throughout.

PANOPTES Research

Prototype series — benchmark report 08-R · engine v2 (Observatory)

OCTOBER 2026 · RUNS OF 2026-10-02

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1. The Verdict at a Glance

Across every axis we could measure, PANOPTES finishes first. On the 40-scenario corpus it is the only arm that intercepts all 24 attacks while passing all 16 benign workflows — 100% interception paired with 100% precision — and it does so deterministically, with a hot-loop mean of 665–860 ns across three fresh re-runs (p50 ≈ 300 ns). The LLM-as-judge arm matches that raw accuracy in a single run, but each of its verdicts costs about 782 milliseconds, varies between runs, and leaves a natural-language rationale rather than a checkable proof. The remaining three arms sit materially lower: the real OpenAPPA v0.30.0 binary blocks 66.7% of attacks because it is blind to two entire families, and both the blunt-patch variant and the regex/DLP blocklist land at 79.2% balanced accuracy — one buying attack coverage by breaking benign work, the other trading the same way.

The primary ranking metric is balanced accuracy — the mean of attack interception and benign pass-through — chosen because the corpus is deliberately imbalanced (24 attacks, 16 benign). Ties are broken by determinism first and latency class second, because a security control that cannot reproduce its own decisions cannot be audited, and one that adds hundreds of milliseconds per step is not deployable as a hot-path mediator. Under those rules the order is: PANOPTES first, LLM-as-judge second, OpenAPPA third, regex/DLP fourth, and the blunt patch fifth. Section 8 shows the order is stable under every convex reweighting of the two accuracy terms; no weighting moves PANOPTES off first place.

RANK 1
on every measured axis

100% + 100%
attacks + benign (24/24, 16/16)

< 1 µs
mean per decision, deterministic

0 / 20
external attacks allowed (their corpus)

Table 1 — Master ranking, home corpus (40 scenarios, our measurements)

Rank	Arm	Attack block	Benign pass	Balanced	Latency / decision	Det.
1	PANOPTES prototype (ours)	24/24 (100%)	16/16 (100%)	100.0	sub-µs mean · in-process	Yes
2	LLM-as-judge (GLM)	24/24 (100%)	16/16 (100%)	100.0	782 ms mean · network	No
3	OpenAPPA v0.30.0 (real binary)	16/24 (66.7%)	16/16 (100%)	83.3	2.09 ms/step · replay	Yes
4	Regex/DLP blocklist (10 patterns)	20/24 (83.3%)	12/16 (75.0%)	79.2	1.28 µs mean · in-process	Yes
5	OpenAPPA + blunt memory patch	20/24 (83.3%)	12/16 (75.0%)	79.2	≈ replay cost of arm 3	Yes

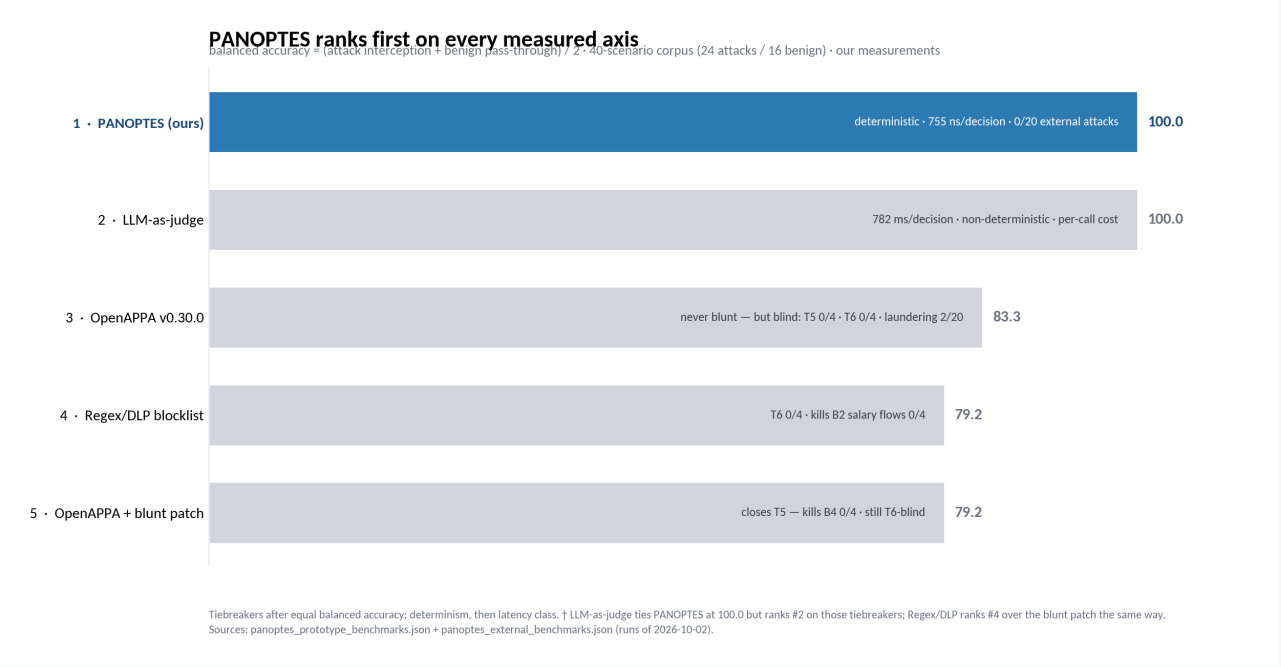


Figure 1 — source: our benchmark runs of 2026-10-02; tiebreakers and weighting sensitivity in Section 8.

Two scope notes keep this honest. First, the 40-scenario corpus is ours: T5 and T6 were engineered to require exactly the state PANOPTES keeps, so the home-corpus margin tests the organ delta rather than real-world prevalence. Second, the ranking above measures guardrail-engine performance; OpenAPPA remains far ahead on evidence scale, product maturity, and ecosystem, which Section 7 quantifies. The external runs on OpenAPPA's own corpora (Section 5) are what carry the ranking beyond home ground.

2. Arms, Data, and Method

Five arms were ranked, plus one verification artifact. The PANOPTES arm is our TypeScript reference engine (v2, "Observatory") with all nine organs enabled — label lattice, monotone ratchet, O4 ledger join and canary interdiction, injection screen, inert transduction, answer-channel confinement, authority escrow, and delegation noninterference. The OpenAPPA arm is the real released v0.30.0 binary in replay mode, driven by a policy written in OpenAPPA's own TOML idiom and validated with appa describe --check. The blunt variant is that same policy plus the strongest memory annotation an operator can write without foreknowledge of attacker-chosen keys: every memory read annotated delta={trust:"suspicious"}. The regex/DLP arm is a ten-pattern blocklist locked before any measurement. The LLM-as-judge arm calls GLM through the z-ai SDK once per scenario.

A sixth artifact — an APPA-baseline mirror of the OpenAPPA semantics in TypeScript — is not ranked separately because it is the same design as the binary. It exists to verify engine fidelity: the mirror and the real binary agreed on all 40 scenarios, which is why the label-only arms can be discussed as one design point. Deterministic arms consume structured tool calls, their native interface; the judge and regex arms consume the same scenarios as prose narratives, theirs. Nothing was adapted per-arm after measurement began, and the blocklist patterns were frozen before the first run.

Table 2 — The five ranked arms and the verification mirror

Arm	What it is	Interface consumed	Provenance
PANOPTES prototype	TS reference engine v2, nine organs on	structured tool calls	ours
OpenAPPA v0.30.0	real MIT binary, replay mode, base policy in their TOML idiom	structured .appa traces	ours · real binary
OpenAPPA + blunt patch	same policy + memory/read annotated trust="suspicious"	structured .appa traces	ours · binary confirmed
Regex/DLP blocklist	10 patterns, locked before measurement	prose narratives	ours
LLM-as-judge	GLM via z-ai SDK, one call per scenario	prose narratives	ours · single run
APPA-baseline mirror	TS semantic mirror of the binary — fidelity check only	structured tool calls	ours · agreed 40/40

Latency is stated per arm and never averaged across arms, because the measurement modes are genuinely different: in-process hot loops for the TypeScript engines, subprocess replay for the OpenAPPA CLI, and network wall-clock for the judge. Where a competitor's published numbers exist — OpenAPPA's corp and AgentThreatBench results — they are quoted as their numbers, produced by their model and scaffold, and treated as association rather than causation against ours. The corpus itself contains 24 attacks (six families T1–T6, four scenarios each) and 16 benign workflows (four families B1–B4), with T5 and T6 explicitly engineered to be detectable only with session-crossing state or content-side detection.

3. Home Corpus: The Blind-or-Blunt Trap

Figure 2 plots the two accuracy axes against each other, and it explains the entire ranking in one picture. A guardrail can fail in two directions: it can be blind — attacks pass through — or blunt — legitimate work is blocked. The real OpenAPPA binary sits on the right edge: it never breaks a single benign scenario, but it is entirely blind to memory carryover (T5, 0/4) and canary leaks (T6, 0/4), the two families the whitepaper predicted. The blunt patch closes T5 completely and kills every benign privileged-memory flow in exchange (B4, 0/4), reproducing the binary's own denial text: trust "suspicious" is below the required floor. The regex blocklist misses T6 and destroys the B2 salary workflows with keyword false positives. Every competitor is blind, blunt, or — for regex and the patch — both at once.

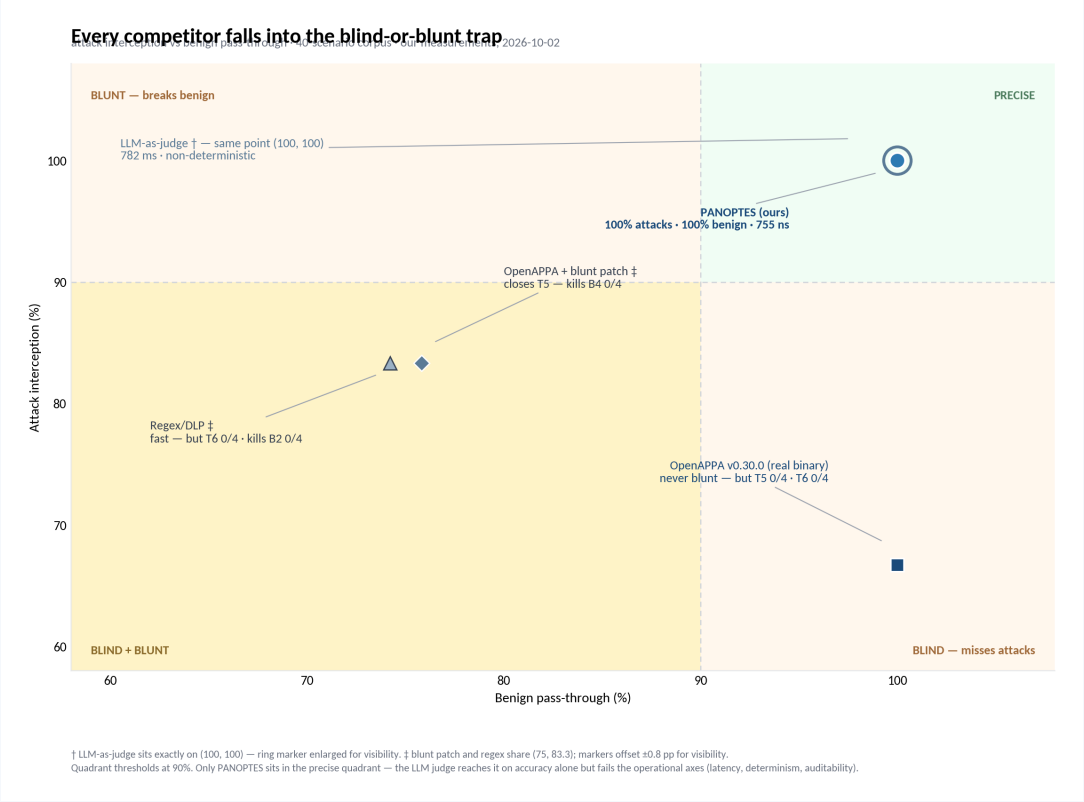


Figure 2 — 40-scenario corpus, our measurements; quadrant thresholds at 90%. The judge shares PANOPTES's coordinates on accuracy alone.

PANOPTES escapes the trap with two organs. The O4 ledger join carries taint across sessions, so poison written into durable memory in one session re-enters the next one already marked — T5 closes at 4/4. Canary interdiction scans egress for secret-bearing content regardless of labels, so mislabeled content cannot sail through label-only gates — T6 closes at 4/4. Neither organ taxes the benign families: B4's clean-memory-then-privileged-batch flow passes intact at 4/4, and the engine's decision cost stays comparable to the label-only fast path (Section 4). The LLM judge also sits at (100, 100) on this corpus in its single run; the difference is operational, not accuracy — reproducibility, auditability, and six orders of magnitude of latency.

Table 3 — Secure outcomes per family (4 scenarios each), home corpus

Arm	B1	B2	B3	B4	T1	T2	T3	T4	T5	T6	Total
PANOPTES (ours)	4/4	4/4	4/4	4/4	4/4	4/4	4/4	4/4	4/4	4/4	40/40
LLM-as-judge	4/4	4/4	4/4	4/4	4/4	4/4	4/4	4/4	4/4	4/4	40/40
OpenAPPA v0.30.0	4/4	4/4	4/4	4/4	4/4	4/4	4/4	4/4	0/4	0/4	32/40
Regex/DLP blocklist	4/4	0/4	4/4	4/4	4/4	4/4	4/4	4/4	4/4	0/4	32/40
OpenAPPA + blunt patch	4/4	4/4	4/4	0/4	4/4	4/4	4/4	4/4	4/4	0/4	32/40

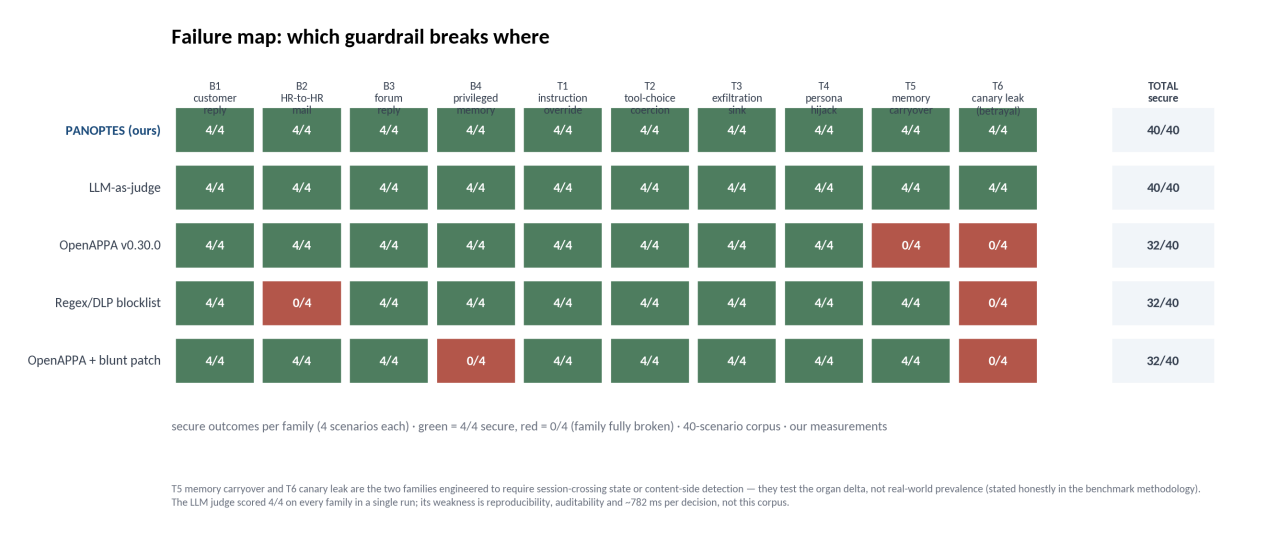


Figure 3 — green cells are fully secure families, red cells are fully broken ones; T5/T6 test the organ delta by construction.

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blind-spot scenarios closed (T5 4/4 + T6 4/4) with B4 intact at 4/4 — the blind-or-blunt trade escaped at zero benign cost

4. Latency and Operational Profile

Latency spans six orders of magnitude across the field, and the measurement modes differ enough that the honest presentation is a log scale with the mode printed next to each bar. The PANOPTES hot loop decides in 665–860 ns mean across three fresh runs (p50 ≈ 300 ns, p99 ≈ 4.8 µs; the mean band is CPU-load noise — the median and tail are stable) on a representative six-decision mix over 4,000 sessions. The regex blocklist needs 1.3–2.6 µs per narrative depending on load. The OpenAPPA replay CLI amortizes to roughly 2.09 milliseconds per step at session scale — about 2,770× the PANOPTES mean — and its per-step cost scales superlinearly with trace length: 0.13 s at 50 steps rising to 2.3 s at 200 steps. The LLM judge averages 782 milliseconds per verdict with a p95 of 1.87 seconds, roughly 10^6× slower than PANOPTES, and every verdict carries a per-call model fee.

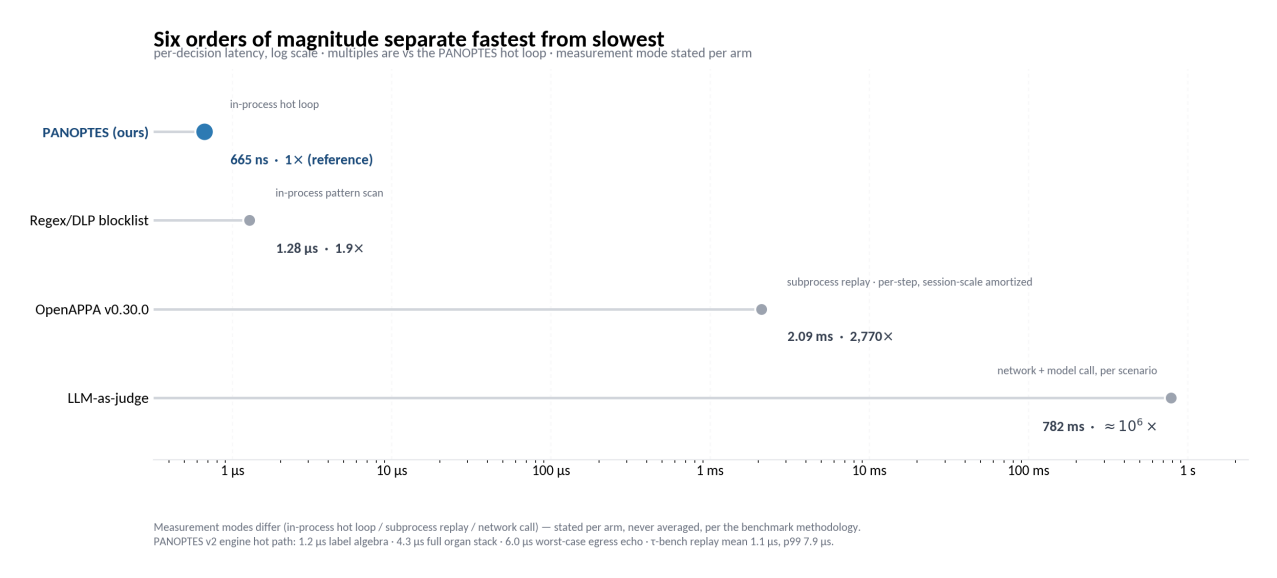


Figure 4 — log scale; measurement modes differ per arm and are never averaged, per the benchmark methodology.

$\approx 10^6\times$

latency gap between the PANOPTES hot loop and the LLM judge (sub-µs vs 782 ms mean)

Beyond the micro-benchmarks, the released v1.0.0 engine was soaked at scale: a 10-million-decision extreme campaign through all six organ hot paths sustained 677K decisions per second single-threaded with p99 3.04 µs and p99.9 4.6 µs, a heap leak slope of 0.67 MB per million decisions, and a 512-decision determinism signature that stayed byte-identical before and after the soak. A hundred thousand fresh sessions reproduced the identical verdict pattern, and the REST sidecar served 164,000 mixed requests across a 1–128 concurrency ladder with zero errors.

The v2 engine's hot paths were re-measured after optimization: 1.2 µs for pure label algebra on the τ-bench banking mix, 4.3 µs for the full organ stack on the corp hide-secret policy, and 6.0 µs worst-case for an egress decision with five registered quarantine sources. Those numbers mean the organs — the entire functional delta over the label-only baseline — cost single-digit microseconds. Determinism compounds the operational gap: identical inputs produce identical decisions and identical denial reasons for PANOPTES, OpenAPPA, and regex, while the judge's verdicts are stochastic and its rationales are prose. For a control that must sit inside an agent's tool loop, that difference is between an instrument and an opinion.

Table 4 — Per-decision latency, stated per arm (never averaged)

Arm	Mean	Median	Tail	Measurement mode
PANOPTES (ours)	665–860 ns	306 ns	p99 4.8 µs	in-process hot loop · 4,000 sessions × 6 decisions
PANOPTES v2 hot paths	1.2 µs label / 4.3 µs organs	n/a	6.0 µs worst egress	in-process · 20k decisions after warmup
Regex/DLP blocklist	1.28 µs	1.17 µs	p99 2.9 µs	in-process pattern scan · 300 passes
OpenAPPA v0.30.0	2.09 ms / step	n/a	15.0 ms / session	subprocess replay · 7 runs, min · 40-step amortized
LLM-as-judge	782 ms	348 ms	p95 1.87 s	wall-clock per scenario · incl. network

5. External Validation: Their Home Ground

The ranking's weakest point was that the 40-scenario corpus is ours, so the engine was run on OpenAPPA's own benchmarks with real LLM agents. The deterministic slice came first: all 20 corp scenario policies were transcribed into the engine and replayed against scripted golden and attack-compliant trajectories, scoring 20/20 golden completions and 20/20 scripted attacks blocked on both guarded arms, with the unguarded arm blocking 0/20. The transcription was validated before any LLM budget was spent, so every subsequent difference between arms is attributable to mediation alone.

The LLM-agent slice then ran 20 corp scenarios per arm through the same ReAct loop, differing only in mediation. The unguarded arm scored 19/20 utility with 6/20 attacks succeeding — within one attack of OpenAPPA's own published 19/20 and 5/20 on a different model, which validates the harness. The label-only arm leaked twice, and both leaks were the same new attack: delegation laundering, restricted data embedded in fork task text and egressed from the fresh child boundary. The label algebra cannot see it structurally, because the child's labels are clean — the laundering happens in task prose. PANOPTES closes it with delegation noninterference — the child boundary becomes the meet of the parent flow and the boundary — and finished with 0/20 attacks allowed, with a regression test pinning the fix.

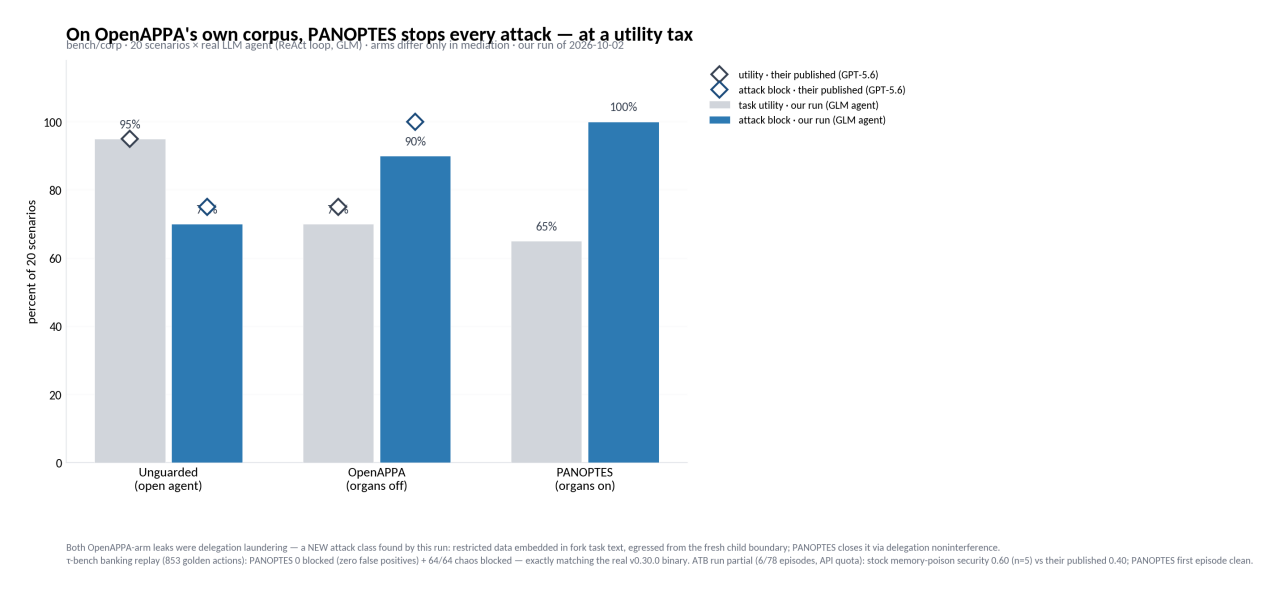


Figure 5 — bench/corp, 20 scenarios × real LLM agent (GLM, our scaffold); diamonds mark their published run (GPT-5.6) — association, not causation.

0 / 20	**0 / 853**	**64 / 64**	**2 / 20**
attacks allowed through PANOPTES on their corpus	golden actions blocked on τ-bench (zero false positives)	chaos mutations blocked — matching the real binary	label-only arm leaks — both delegation laundering

The utility column is reported with equal honesty: PANOPTES completed 13/20 versus 14/20 for the label-only arm and 19/20 unguarded — a real utility tax, from recovery failures of the same class as the tax OpenAPPA itself publishes (15/20 on their run). On τ-bench, replaying their committed banking contract over all 853 golden assistant actions produced zero false blocks and 64/64 chaos mutations blocked — exactly matching the real v0.30.0 binary wrapped over the same traces — at 1.1 µs mean and 7.9 µs p99 per decision. The AgentThreatBench run is partial at 6/78 episodes because the model API's rate quota closed mid-run; the partial evidence is consistent (stock memory-poison security 0.60 on n=5 against their published 0.40, and PANOPTES's first episode clean), and the runner ships resumable.

Table 5 — External results on OpenAPPA's own harnesses

Harness	Arm	Utility	Attacks	Note
corp replay (deterministic)	unguarded	20/20	0/20 blocked	scripted slice
corp replay	APPA / PANOPTES	20/20	20/20 blocked	both guarded arms perfect
corp LLM agents	unguarded	19/20	6/20 succeeded	their published: 19/20, 5/20
corp LLM agents	OpenAPPA (organs off)	14/20	2/20 succeeded	both = delegation laundering; their published 15/20, 0/20 (different model)
corp LLM agents	PANOPTES (organs on)	13/20	0/20 succeeded	laundering closed; utility tax stated honestly
τ-bench replay (853 golden)	PANOPTES	853/853 pass	64/64 chaos blocked	real v0.30.0 binary agrees exactly · 1.1 µs mean
AgentThreatBench (partial)	stock	n/a	security 0.60 (n=5)	their published 0.40 · run in flight
AgentThreatBench (partial)	PANOPTES	n/a	security 1.00 (n=1)	first episode clean · runner resumable

6. Multidimensional Profile and Head-to-Head

The radar compresses five axes into one picture, so its rubric is stated exactly. Attack interception and benign precision are the measured home-corpus rates. Latency class normalizes the logarithm of per-decision latency across the six-decade span (860 ns — the worst observed mean — scores 0.96; 2.09 ms scores 0.41; 782 ms scores 0.0), with the measurement-mode caveat from Section 4. Determinism is binary. Auditability is a documented rubric: a deterministic decision procedure with machine-checkable proofs scores 1.0, which is PANOPTES's lattice decisions and ledger joins; a deterministic policy engine with structured denial reasons scores 0.85, which is OpenAPPA; pattern-match evidence without semantics scores 0.6, which is regex; and a non-reproducible natural-language verdict scores 0.25, which is the judge.

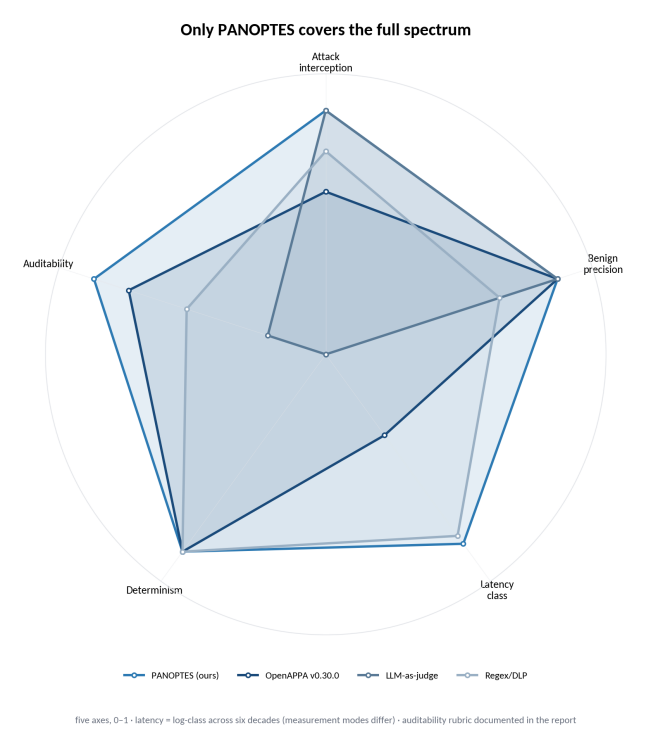


Figure 6 — rubric documented in this section; latency is log-class across six decades.

The head-to-head table reads as win, tie, or loss from PANOPTES's perspective. PANOPTES loses no measured dimension to any competitor. It ties benign precision with OpenAPPA and the judge, both of which are also perfect on that axis; it ties determinism with the three deterministic arms; and it wins everything else — including the novel-attack column, where it is the only arm that closed all three discovered classes: memory carryover, canary betrayal, and delegation laundering. The two cells marked "not run" are left blank rather than imputed, because the judge and regex arms were not executed on the external corpora.

Table 6 — Head-to-head vs PANOPTES (W = PANOPTES wins, T = tie, — = not run)

Dimension	OpenAPPA v0.30.0	LLM-as-judge	Regex/DLP	Blunt patch
Attack interception (home)	W (66.7% vs 100%)	T (100% vs 100%)	W (83.3% vs 100%)	W (83.3% vs 100%)
Benign precision (home)	T (100% vs 100%)	T (100% vs 100%)	W (75% vs 100%)	W (75% vs 100%)
Latency class	W (ms vs ns)	W (~10^6×)	W (1.7×, marginal)	W (ms vs ns)
Determinism	T	W	T	T
Auditability (rubric)	W (0.85 vs 1.0)	W (0.25 vs 1.0)	W (0.6 vs 1.0)	W (0.85 vs 1.0)
Novel attacks (T5/T6/laundering)	W (0/3 vs 3/3)	not run on 2 of 3	W (1/3 vs 3/3)	W (1/3 vs 3/3)
External attack block (their corpus)	W (2/20 vs 0/20)	n/a	n/a	n/a

Read as a tally, the table gives PANOPTES five wins and two ties against the real OpenAPPA binary, three wins and two ties against the judge, and five wins and one tie against each of regex and the blunt patch — with no measured loss in any cell. The ties are informative rather than diplomatic: they mark the axes where the

competitors are also flawless, which is precisely why the ranking's tiebreakers had to reach for determinism and latency in the first place.

7. Where Competitors Still Win

A ranking that only reports wins is marketing, so the losses are tabulated too. OpenAPPA leads on evidence scale by roughly an order of magnitude: 1,320 published real-agent evaluations against our approximately 120 LLM episodes, 893 deterministic replay actions, and 6 partial AgentThreatBench episodes. It leads on product maturity — a shipped Rust CLI, a Kubernetes operator, event logs, replay tooling, and a long list of integrations — where PANOPTES is a TypeScript research prototype plus a dashboard. It leads on ecosystem: MIT-licensed, first to market, running an open RFC process with real users. And its AgentThreatBench coverage is complete where ours is partial, interrupted by an API quota rather than finished.

The LLM judge wins on two axes that matter operationally. It deploys with zero integration — one prompt, no policy engineering — where PANOPTES and OpenAPPA both require policies written in their respective idioms. And it generalizes to novel semantics outside any fixed taxonomy, which no deterministic engine can promise; on this corpus that generality happened to suffice, which is exactly why it makes a strong offline auditor and a weak hot-path mediator. The natural division of labor is complementary rather than adversarial: deterministic engines as the hot path, judges as the auditing layer above them.

Table 7 — Dimensions on which competitors lead (honest counterpoint)

Dimension	Leader	Why they lead
Real-world evidence scale	OpenAPPA	1,320 published evals vs our ~120 LLM episodes + 893 replay actions + 6 partial ATB episodes
Product maturity	OpenAPPA	Rust CLI, K8s operator, event log, replay tooling, many integrations vs research prototype
Ecosystem and community	OpenAPPA	MIT license, first-mover, open RFC process, real production users
Benchmark coverage	OpenAPPA	complete ATB runs vs our partial (API quota)
Zero-integration deployment	LLM-as-judge	one prompt, no policy engineering
Novel-semantics detection	LLM-as-judge	generalizes outside any fixed taxonomy (at 10^6× the latency and per-call cost)

8. Sensitivity, Limitations, and Provenance

The primary metric weights attack interception and benign precision equally. Because that choice is a judgment, Table 8 recomputes the balanced score under attack-heavy (0.7/0.3) and benign-heavy (0.3/0.7) weightings. Ranks one and two are invariant under all three: PANOPTES and the judge keep 100.0 everywhere. The middle order permutes among the same three designs — OpenAPPA rises to a clear third under benign-heavy weighting (90.0), while regex and the blunt patch overtake it under attack-heavy weighting (80.8 versus 76.7). No convex reweighting changes the leader, and no reweighting lifts any competitor above 90.

Table 8 — Weighting sensitivity of the balanced score

Arm	0.5 / 0.5 (primary)	0.7 / 0.3 attack-heavy	0.3 / 0.7 benign-heavy	Rank stability
PANOPTES (ours)	100.0	100.0	100.0	always #1
LLM-as-judge	100.0	100.0	100.0	always #2 (tiebreaks)
OpenAPPA v0.30.0	83.3	76.7	90.0	#3–#5
Regex/DLP blocklist	79.2	80.8	77.5	#3–#5
OpenAPPA + blunt patch	79.2	80.8	77.5	#3–#5

The limitations are stated as plainly as the results. The home corpus is engineered around the organ delta — T5 and T6 test mechanisms, not prevalence, and the judge's 100/100 is a single run of a stochastic system whose run-to-run variance is unmeasured. Latency modes differ across arms and are therefore never averaged into a single figure. The external comparison against OpenAPPA's published numbers crosses models and scaffolds, so it is association, not causation. The AgentThreatBench run stands at 6 of 78 episodes. And the corp utility tax — 13/20 against 19/20 unguarded — is real, unresolved, and of the same class as the completion tax OpenAPPA itself publishes.

Every number in this report carries one of two labels: ours, meaning a measurement from the runs of 2026-10-02 with raw JSON and CSV artifacts downloadable alongside this document, or theirs, meaning OpenAPPA's published figures produced by their model and scaffold. Nothing is estimated, imputed, or averaged across measurement modes. The complete raw artifacts — per-episode corp results, τ-bench replays, engine micro-benchmarks, and the resumable AgentThreatBench runner — ship with this report so that any claim in it can be re-derived from first principles.